1.Project Idea / Concept:

Project Name : Jobify -A robust recruitment platform

- **Problem Statement:** Many job seekers, especially students and fresh graduates, struggle to quickly find suitable job opportunities. Existing platforms are often complex, overloaded with irrelevant listings, or not focused on early-career candidates.

- **Proposed Solution:** A web-based recruitment platform that connects job seekers with employers simply and efficiently. The system allows companies to post job openings and enables candidates to search for jobs, upload CVs, and apply directly through the platform.

- **Target Audience:** Students, fresh graduates, job seekers, startups, and companies looking to hire qualified candidates.

- **Unique Value Proposition (UVP):** A clean and easy-to-use platform with **AI-assisted job matching**, where the system analyzes the candidate's CV and recommends suitable jobs based on skills and experience.

- **Core Features:**

    - Role-Based Authentication (Admin, Employer, Job Seeker).
    - Job Posting and Management for Employers.
    - Job Search and Filtering by category, location, or skills.
    - CV Upload and Job Application System.
    - Admin Dashboard for managing users, jobs, and reports.

2.Team Structure:

Team Size: 5 Members

Ahmed Hany: Team Leader / Software Architect

Role: Sets up the GitHub repository, defines the N-Tier Architecture (Web, Services, Data layers), and reviews pull requests.

Abdulrahman Hany: Back-End Developer (Business Logic & Payments)

Role: Writes the C# business logic in the Services layer, calculates cart totals, and integrates the Stripe Payment Gateway.

Youssif Othman: Database Architect (Data Layer)

Role: Designs the SQL Server database, writes Entity Framework Core configurations, and implements the Repository & Unit of Work patterns.

Abdulrahman Ashraf: Front-End Developer (Customer UI& Admin Dashboard)

Role: Designs the user-facing storefront using Bootstrap, HTML5, and CSS3. Ensures the website is fully responsive on mobile phones.

Builds the Admin panel using Query and Data Tables to allow sorting, searching, and Pagination for managing products and viewing orders

Dalal Mohamed: QA Tester (Quality Assurance)
Role: Tests the platform to ensure all features work correctly and the system is free of bugs. Verifies job posting, job applications, user registration, and login functions. Performs functional and usability testing to ensure a smooth user experience and reports issues to the development team.

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3. System Analysis

The Blueprint of the Job Recruitment Platform

Functional Requirements

- Job seekers must be able to **create accounts, log in, and manage their profiles**.
- Job seekers must be able to **upload resumes and update their skills and experience**.
- Users must be able to **search for jobs by title, category, location, or required skills**.
- Job seekers must be able to **apply for jobs directly through the platform**.
- Companies must be able to **create company accounts and post job vacancies**.
- Companies must be able to **view applications submitted by candidates**.
- Companies must be able to **accept, reject, or review applications**.
- The system must send **notifications when a job application is submitted or reviewed**.

Non-Functional Requirements

Security:

- User passwords must be **encrypted/hashed before storing in the database**.
- Only authorized users (companies) can **create or manage job postings**.
- The platform must use **secure authentication and protected APIs**.

Performance:

- Job search results must **be loaded within 2 seconds**.
- The platform should support **multiple users by accessing the system simultaneously**.

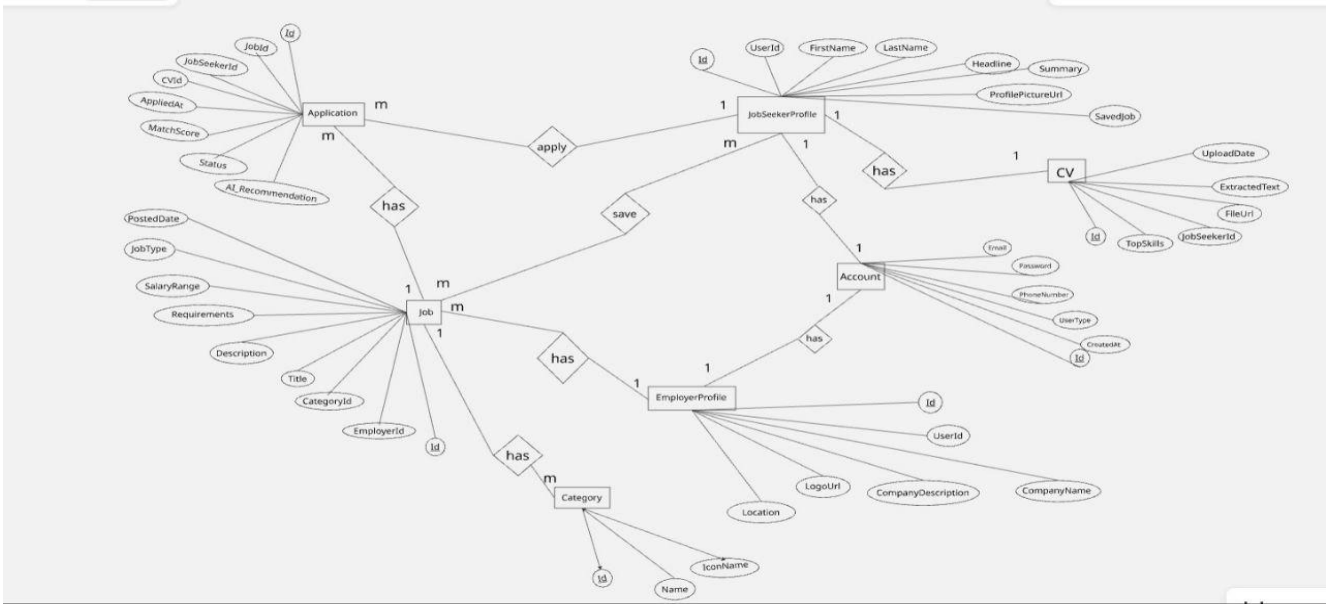
Usability:

- The interface must be **simple and responsive on mobile and desktop devices**.

Database Design (ERD Entities):

The **ERD** illustrates the database structure and relationships between the main entities of the system. Key entities include **Account, Jobseeker Profile, Employer Profile, Job, Application, CV, and Category**.

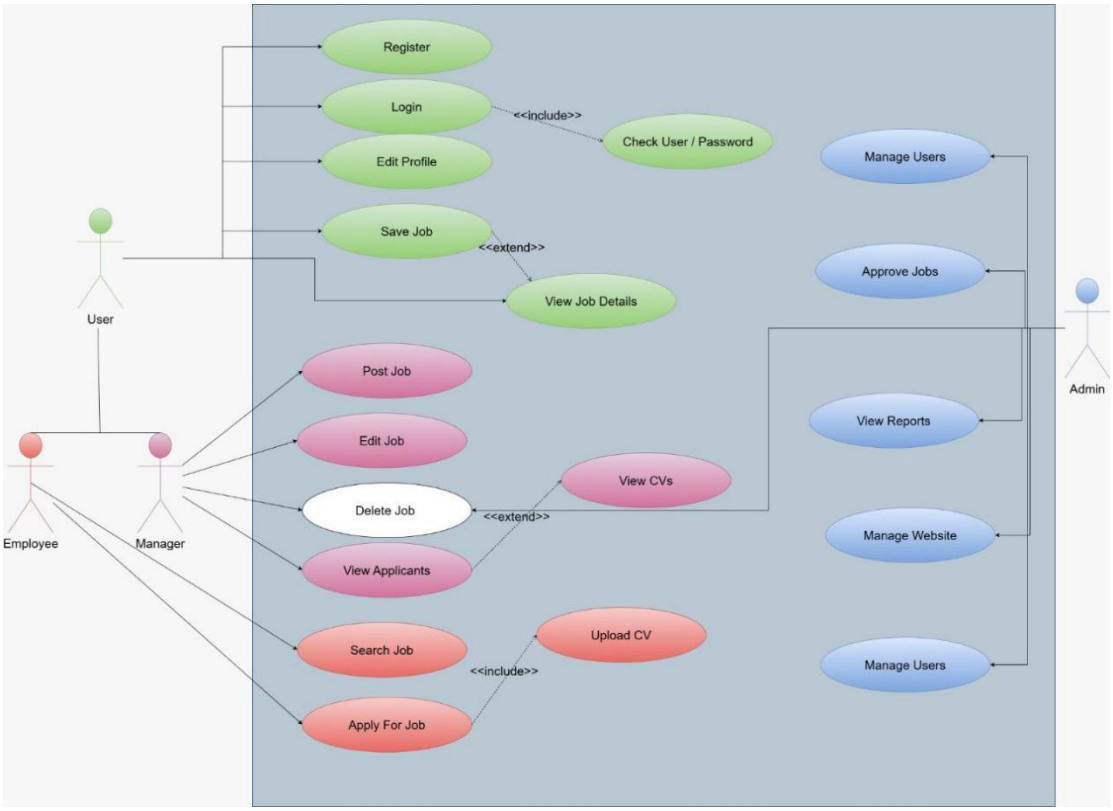
A **job seeker** can apply to multiple jobs, creating **applications**, while each **job** can receive many applications.
Jobs belong to **categories**, and each **account** can have either a **job seeker profile** or an **employer profile**.
Job seekers can also **upload CVs**, which are stored and linked to their profiles.



Use Case:

The **Use Case Diagram** shows how different actors interact with the job recruitment platform. The main actors are **User (Job Seeker), Employer (Employee/Manager), and Admin**.

A **User** can register, log in, edit their profile, search for jobs, view job details, save jobs, upload a CV, and apply for jobs.
An **Employer** can post, edit, and delete job postings and view applicants and their CVs.
The **Admin** manages the system by approving jobs, managing users, viewing reports, and controlling the website.



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4. Case Study / Feasibility Study
Technical Feasibility
We have the necessary technical expertise and a defined technology stack to build Jobify successfully.

- **Architecture**: Using an **N-Tier Architecture** (Web, Services, Data layers) allows the team to work on separate concerns simultaneously, reducing development bottlenecks.

- **Back-End Proficiency**: Our team includes developers skilled in **C# business logic** and **Entity Framework Core**, ensuring the system can handle complex database relationships and external integrations like **Stripe**.
- **Front-End Capabilities**: We are utilizing **Bootstrap, HTML5, and CSS3** to ensure the platform is responsive across mobile and desktop devices.
- **AI Integration**: The feasibility of our AI-assisted job matching rests on implementing text extraction from uploaded CVs to compare candidate skills against job requirements.

Economic Feasibility

A cost-benefit analysis indicates that Jobify is a viable investment of time and resources.

- **Development Costs**: Minimal, as the team is using open-source frameworks (.NET Core, SQL Server, Bootstrap) and existing hardware.
- **Operational Benefits**: The platform automates the recruitment process, significantly reducing the time students and graduates spend searching for relevant entry-level roles.
- **Scalability**: The modular N-Tier design allows for future monetization strategies, such as premium employer listings or advanced AI analytics, without requiring full system rewrite.

Operational Feasibility

The system is designed specifically to solve the usability issues of current job platforms.

- **Ease of Use**: By focusing on a **clean and simple UI**, we ensure that students and fresh graduates—who are often overwhelmed by complex platforms—can navigate the system with ease.
- **Defined Workflows**: With **Role-Based Authentication** (Admin, Employer, Job Seeker), users are only presented with the tools they need, reducing confusion and increasing system adoption.
- **Quality Assurance**: A dedicated **QA Tester** will perform functional and usability testing to ensure the system is bug-free and user-friendly before deployment.

Competitor Analysis (Case Study)

Competitor	Strengths	Weaknesses	Jobify's "Gap" Solution
LinkedIn	High professional networking value.	Overloaded with irrelevant listings for fresh graduates; complex interface.	**Specialized Focus**: Dedicated exclusively to early-career candidates and startups.
Indeed	Huge volume of data and international reach.	Generic matching often requires manual filtering through thousands of non-relevant jobs.	**AI-Assisted Matching**: Automatically recommends jobs by analyzing the candidate's CV and skills.
Wuzzuf	Strong local presence in the Egyptian market.	UI can feel cluttered; profile setup is time-consuming for first-time users.	**Streamlined Experience**: Simple CV upload and direct application system to get users hired faster.

Risk Assessment Matrix

Potential Risk	Impact	Probability	Mitigation Strategy

Data Privacy (CVs/Personal Info)	High	Medium	Implement password hashing and secure, protected APIs for all profile data.
System Performance/Latency	Medium	Medium	Optimize SQL queries and EF Core configurations to ensure search results load within 2 seconds.
AI Matching Inaccuracy	Medium	High	Use the QA process to verify that the extracted text accurately reflects the candidate's "Top Skills."
User Unauthorized Access	High	Low	Enforce strict Role-Based Authentication to prevent Job Seekers from accessing Admin or Employer tools.